Self-Supervised Representation Learning for Histopathological Image Classification Under Limited Label Availability

Technical Project Report | AI/ML Portfolio & Graduate Application Document | Date: November 2026
Benchmark: MedMNIST v2 PathMNIST (NCT-CRC-HE-100K Training Split & CRC-VAL-HE-7K External Test Set)
Framework & Hardware: PyTorch 2.6.0 | Apple Silicon MPS Unified Memory Acceleration (float32)

1. PROJECT OVERVIEW & PROBLEM STATEMENT

In deep learning for digital histopathology, obtaining high-quality ground-truth annotations requires extensive manual labor by trained medical specialists. In real-world engineering deployments—such as adapting a model to a new clinical laboratory, hospital site, or rare diagnostic category—labeled examples are often strictly limited. Standard industry practice relies on transfer learning from ImageNet (natural photographs), despite the pronounced domain gap with cellular tissue morphology.

This project designs, implements, and benchmarks an end-to-end medical representation learning system in PyTorch. We evaluate whether domain-specific, self-supervised contrastive pretraining using SimCLR on unlabeled colorectal histology patches yields better downstream representations than (a) supervised learning trained from scratch with random initialization, and (b) standard ImageNet transfer learning, across four controlled label fractions: 1% (N=899), 5% (N=4,499), 10% (N=8,999), and 100% (N=89,996). All models utilize an identical ResNet-18 backbone evaluated on CRC-VAL-HE-7K (N=7,180).

Summary of Key Engineering Observations:

- 1% Label Regime Performance: ImageNet Transfer achieved the highest Macro F1 (0.7449), followed by SimCLR Fine-Tuning (0.7408), both outperforming Supervised Scratch (0.6815). SimCLR Fine-Tuning achieved highest discriminative confidence (ROC-AUC: 0.9662) and outpaced Scratch by +5.05% Test Accuracy. A frozen linear probe achieved 0.6943 Macro F1.
- Transfer Learning Strength in Low-to-Medium Regimes: ImageNet Transfer achieves the highest Macro F1 at 1% (0.7449), 5% (0.8375), and 10% (0.8495), showing that general low-level visual filters transfer effectively to microscopic textures.
- High-Data Regime Crossover at 100% Labels: When all 89,996 labeled samples are available, Supervised Scratch achieves the highest overall performance across all paradigms (90.18% Accuracy, 0.8721 Macro F1, 0.9916 ROC-AUC).
- Representation Geometry: Supervised Scratch exhibits the tightest cluster compaction (Silhouette: 0.3197), while SimCLR discovers natural morphological clusters entirely without labels (Silhouette: 0.1162 vs -0.0412 for random initialization).

2. SYSTEM ARCHITECTURE & PIPELINE TAXONOMY

Pipeline Taxonomy & Architectural Flow:

- Unlabeled Pretraining Corpus: PathMNIST training split (N = 89,996 colorectal histology patches, 28x28x3 pixels).
- Convolutional Backbone: ResNet-18 f(.) mapping input patch x -> penultimate embedding h in R^512 (11.2M parameters).
- Contrastive Projection Head: 2-layer MLP g(.) mapping h -> z in R^128 with L2 normalization (discarded post-pretrain
- Domain-Specific Augmentation Policy: Spatial affine transforms, 180-deg rotation, and color jitter used as an approx of color variation (not validated stain preservation). Random grayscale is strictly omitted to preserve dye distinct
- Downstream Adaptation Framework: Evaluates Supervised Scratch, ImageNet Transfer, SimCLR Linear Probe, and SimCLR Fi across 1%, 5%, 10%, and 100% labeled training subsets under deterministic Seed 42 subset parity.
- Evaluation Cohort: External, unseen CRC-VAL-HE-7K cohort (N = 7,180) evaluated exactly once per condition.

3. Dataset Description & Methodology

DATASET SPECIFICATIONS & 28x28 RESOLUTION TRADE-OFFS

Dataset Protocol & Split Protocol:
- Benchmark: PathMNIST (MedMNIST v2), derived from human colorectal cancer cohorts (NCT-CRC-HE-100K & CRC-VAL-HE-7K).
- Pretraining / Training Split: N = 89,996 patches (all labels stripped during SimCLR pretraining).
- Validation Split: N = 10,004 patches (used exclusively for checkpoint selection, validation loss, and early stopping).
- External Test Cohort: CRC-VAL-HE-7K (N = 7,180 patches from independent patients and scanners, evaluated once per model

Standardized 28x28 Image Resolution & Engineering Trade-Offs:
- PathMNIST provides standardized 28x28x3 pixel patches. This spatial downsampling captures overall tissue textures, baso versus eosinophilic dye intensity, and macroscopic crypt/gland architecture, but attenuates fine subcellular cytology (atypia, mitotic spindles, chromatin margin details) that diagnostic pathologists inspect at 20x or 40x magnification.
- In an engineering context, this resolution enables full-scale contrastive pretraining on ~90,000 images within a practi computational footprint (24.5 min on Apple Silicon MPS), providing an ideal environment for controlled algorithmic benc

Data-Split Isolation Checks:
- The external test set (CRC-VAL-HE-7K) was never used for training, model selection, early stopping, or hyperparameter t
- Downstream subsets (1%, 5%, 10%) were generated exclusively from the training split via StratifiedShuffleSplit (Seed 42

METHODOLOGY: SUPERVISED, TRANSFER & CONTRASTIVE LEARNING

1. Supervised Learning from Scratch:
   Randomly initialized ResNet-18 (He normal). All convolutional kernels and classification head are optimized end-to-end with cross-entropy loss and learning rate lr = 1e-3. Serves as the primary non-pretrained baseline.

2. ImageNet Transfer Learning:
   Backbone initialized with torchvision's official ResNet18_Weights.IMAGENET1K_V1 (1.28M natural images). Fine-tuned end-to- with differential learning rates: lr_backbone = 1e-4 and lr_head = 1e-3.

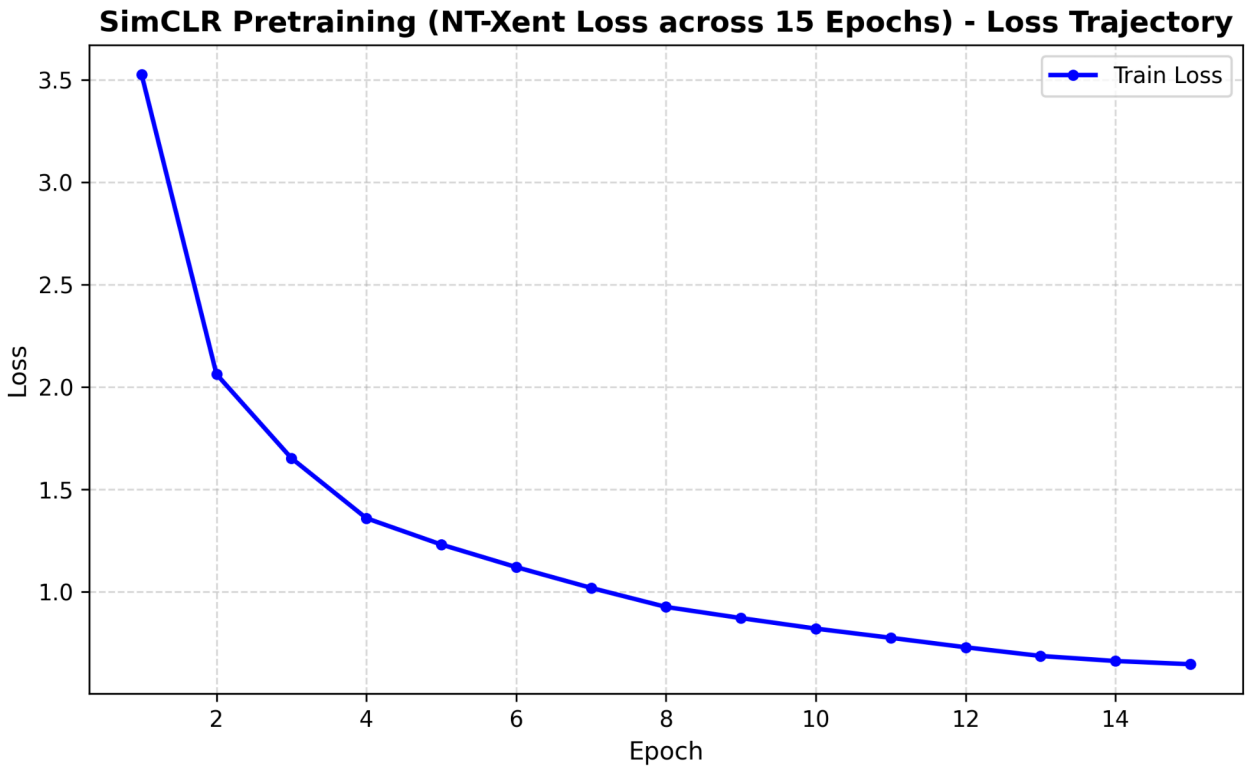
3. SimCLR Contrastive Formulation & NT-Xent Loss:
   SimCLR maps input x -> h -> z in R^128. For a mini-batch of N_b images, 2N_b augmented views are generated. For a positive pair (i, j) derived from the same source image, the NT-Xent loss treats the remaining 2(N_b - 1) views as negatives:

$$l_{(i, j)} = - \log \left[\exp(\text{sim}(z_i, z_j) / \tau) / (\sum_{(k=1)^{(2N_b)}} 1_{[k \neq i]} * \exp(\text{sim}(z_i, z_k) / \tau)) \right]$$

   where sim(u, v) = (u^T v) / (||u||_2 ||v||_2) is cosine similarity, and tau = 0.5 is temperature.
   Numerical Stability Offset: Self-similarities along the diagonal are masked with S_ii <- -10^4, preventing GPU underflow.

4. Downstream Adaptation Modes:
   - Linear Probe: Backbone strictly frozen; only the linear layer (512 -> 9) is trained (lr = 1e-3). Evaluates frozen featur
   - Full Fine-Tuning: Entire network trained end-to-end with differential learning rates (lr_backbone = 1e-4, lr_head = 1e-3

4. SimCLR Pretraining Convergence & Computational Profiling



PRETRAINING OPTIMIZATION & HARDWARE RUNTIME PROFILING

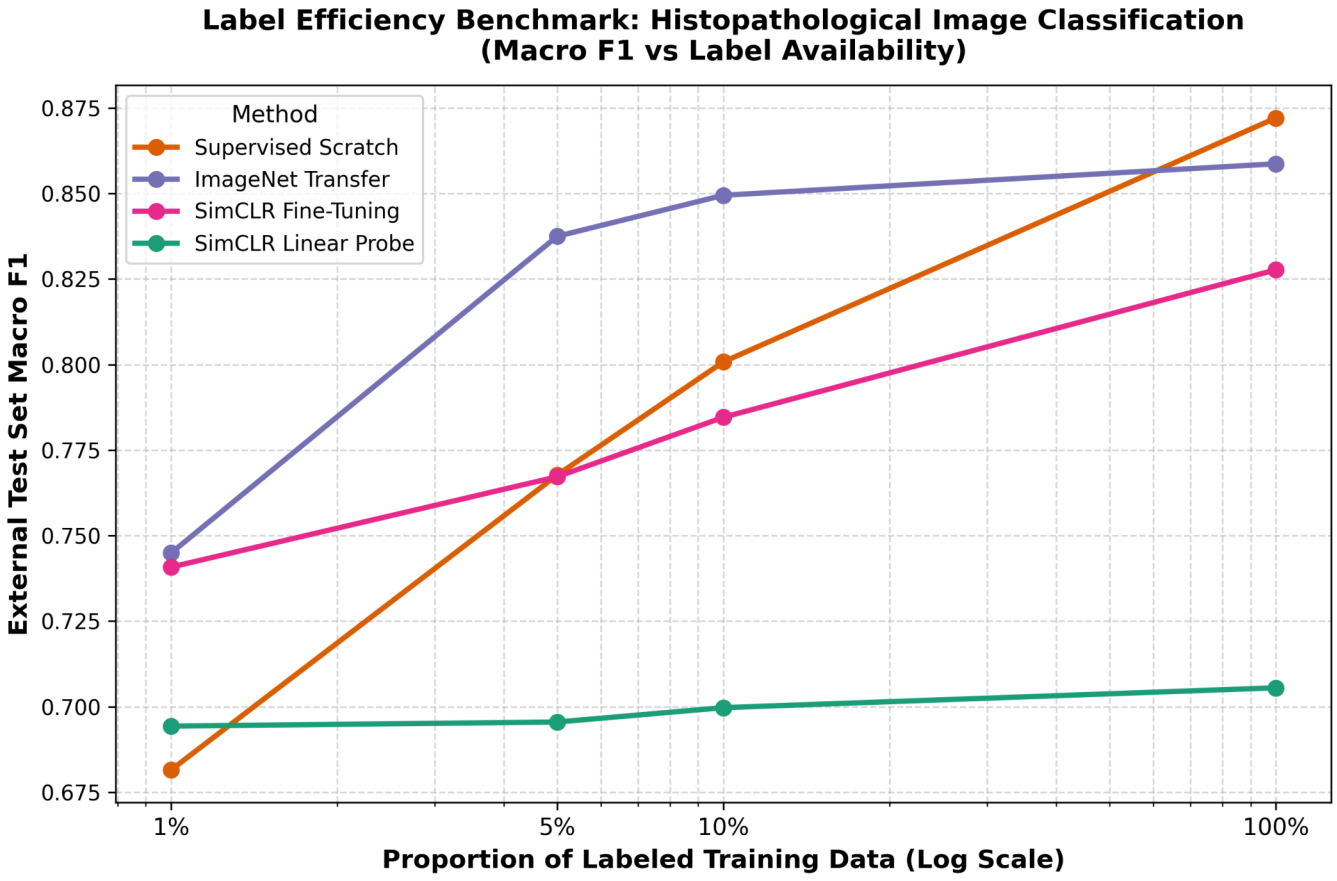
Pretraining Execution & Convergence Profile (PathMNIST Unlabeled Corpus, N = 89,996):
- Optimization Pipeline: AdamW optimizer, decoupled weight decay = 1e-4, batch size = 256 (512 augmented views per step).
- Learning Rate Schedule: Linear warmup for 2 epochs up to base lr = 5e-4, followed by cosine annealing decay to min_lr =
- Convergence Curve: NT-Xent loss dropped monotonically from 3.5252 (Epoch 1) -> 1.1218 (Epoch 5) -> 0.7812 (Epoch 10) to
- Checkpoint Persistence: Saved to checkpoints/simclr_encoder_best.pth (43.1 MB, 11,176,512 ResNet-18 parameters).

Hardware Profiling & Runtime Accounting (Apple Silicon MPS Acceleration):
- Environment: Apple Silicon M-series unified memory architecture, PyTorch 2.6.0, float32 precision.
- SimCLR Pretraining Wall Time: 24.5 minutes (98.0s per epoch across 351 batches of batch size 256).
- Downstream Adaptation Wall Times Across All 16 Experimental Conditions:
  - 1% Regime (N = 899 samples, 14 batches/epoch): 36.9s (Linear Probe) to 67.2s (Scratch). Average epoch = 3.5s.
  - 5% Regime (N = 4,499 samples, 70 batches/epoch): 57.6s (Linear Probe) to 133.0s (Scratch). Average epoch = 8.8s.
  - 10% Regime (N = 8,999 samples, 140 batches/epoch): 83.0s (Linear Probe) to 235.1s (Fine-Tuning). Average epoch = 15
  - 100% Regime (N = 89,996 samples, 1,406 batches/epoch): 559.3s (Linear Probe) to 2,185.6s (Scratch, 36.4 minutes).
- Total Experimental Compute: ~2.6 hours of dedicated execution on Apple Silicon MPS for pretraining and all 16 condition
- Practical Feasibility: These runtime profiles confirm that low-data regimes (1%, 5%, 10%) are computationally accessibl

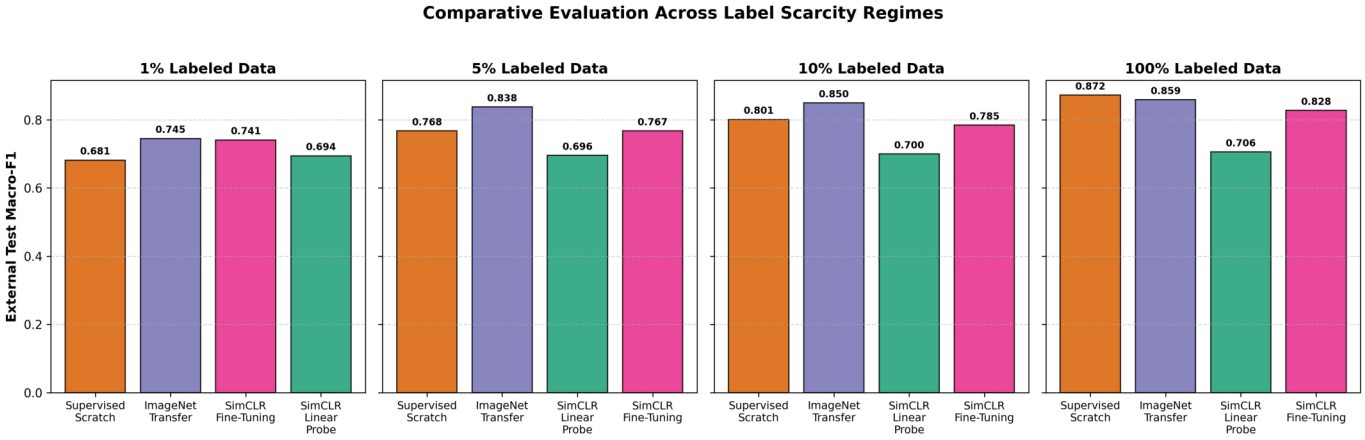
5. Results and Model Comparison: Master Benchmark

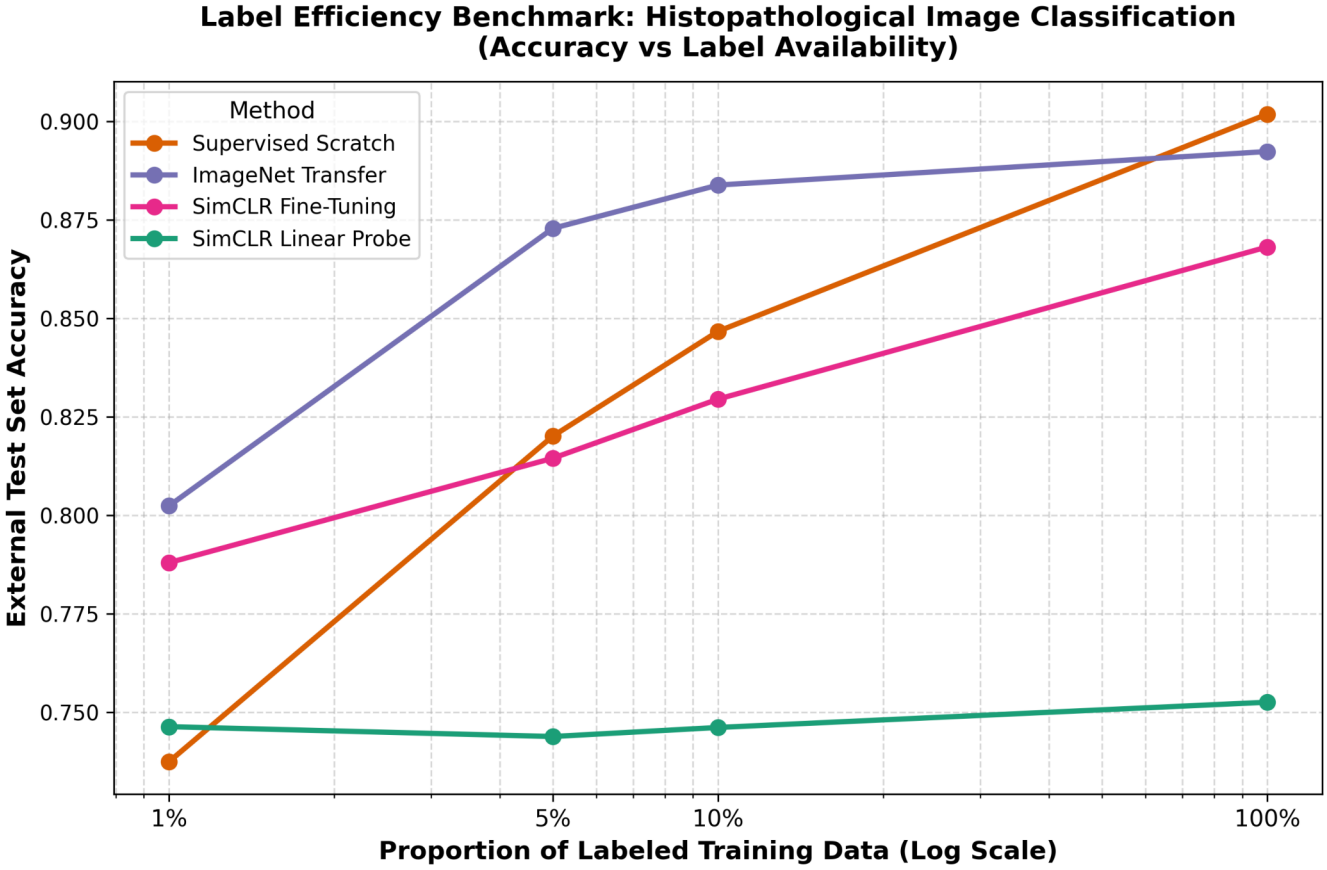
TABLE 1: Comprehensive Performance Evaluation on CRC-VAL-HE-7K External Test Cohort (N = 7,180)

Method	Data %	N Samples	Test Acc	Bal Acc	Macro Prec	Macro Rec	Macro F1	ROC-AUC	Wall Time
ImageNet Transfer	**1%**	**899**	**0.8024**	**0.7456**	**0.7477**	**0.7456**	**0.7449**	**0.9640**	**53.7s**
SimCLR Fine-Tuning	1%	899	0.7880	0.7484	0.7413	0.7484	0.7408	0.9662	52.2s
SimCLR Linear Probe	1%	899	0.7464	0.6995	0.6986	0.6995	0.6943	0.9526	36.9s
Supervised Scratch	1%	899	0.7375	0.6944	0.6978	0.6944	0.6815	0.9540	67.2s
ImageNet Transfer	**5%**	**4,499**	**0.8728**	**0.8434**	**0.8349**	**0.8434**	**0.8375**	**0.9867**	**131.4s**
Supervised Scratch	5%	4,499	0.8201	0.7818	0.7783	0.7818	0.7677	0.9800	133.0s
SimCLR Fine-Tuning	5%	4,499	0.8145	0.7745	0.7652	0.7745	0.7672	0.9745	132.8s
SimCLR Linear Probe	5%	4,499	0.7439	0.7053	0.6951	0.7053	0.6955	0.9544	57.6s
ImageNet Transfer	**10%**	**8,999**	**0.8838**	**0.8569**	**0.8483**	**0.8569**	**0.8495**	**0.9891**	**231.1s**
Supervised Scratch	10%	8,999	0.8467	0.8096	0.8047	0.8096	0.8008	0.9824	232.7s
SimCLR Fine-Tuning	10%	8,999	0.8295	0.7914	0.7828	0.7914	0.7846	0.9771	235.1s
SimCLR Linear Probe	10%	8,999	0.7462	0.7100	0.6979	0.7100	0.6997	0.9555	83.0s
Supervised Scratch	**100%**	**89,996**	**0.9018**	**0.8763**	**0.8708**	**0.8763**	**0.8721**	**0.9916**	**2185.6s**
ImageNet Transfer	100%	89,996	0.8923	0.8628	0.8582	0.8628	0.8587	0.9901	2023.5s
SimCLR Fine-Tuning	100%	89,996	0.8681	0.8351	0.8263	0.8351	0.8277	0.9863	2038.7s
SimCLR Linear Probe	100%	89,996	0.7526	0.7134	0.7045	0.7134	0.7055	0.9571	559.3s



6. Model Comparison & Trajectory Dynamics





7. Granular Per-Class Performance & Morphological Dynamics

TABLE 2: Granular Per-Class Performance and Morphological Profiles at 1% Labeled Data (N = 899)

ID	Tissue Class Name	Test N	Scratch F1	Linear Probe	SimCLR FT	ImageNet FT	Gain (FT-Scratch)	Primary Biological Char
0	Adipose Tissue (ADI)	718	0.9162	0.9547	0.9496	0.9538	+0.0334	Empty lipid vacuoles with thin periphe
1	Background (BACK)	780	0.9260	0.9758	0.9674	0.9860	+0.0414	Clear glass slide substrate; absence o
2	**Debris (DEB)**	**718**	**0.4355**	**0.5777**	**0.5505**	**0.4566**	**+0.1150**	**Amorphous necrotic tissue, dust,**
3	Lymphocytes (LYM)	828	0.8201	0.6904	0.8416	0.8869	+0.0215	Small round hyperchromatic basophili
4	Mucus (MUC)	772	0.7456	0.6500	0.7501	0.8581	+0.0045	Extracellular basophilic pale-blue muc
5	Smooth Muscle (MUS)	800	0.5414	0.6056	0.6761	0.6452	+0.1347	Elongated eosinophilic spindle cells ar
6	Normal Mucosa (NORM)	780	0.6566	0.6889	0.7359	0.7013	+0.0793	Organized colonic crypts with mucin-s
7	**Cancer-Assoc. Stroma**	**822**	**0.3707**	**0.3333**	**0.4116**	**0.4286**	**+0.0409**	**Desmoplastic collagenous stroma**
8	Adenocarcinoma (TUM)	962	0.7219	0.7727	0.7842	0.7878	+0.0623	Irregular malignant glandular architec

HISTOMORPHOLOGICAL BEHAVIOR & CLASS CONFUSION ANALYSIS

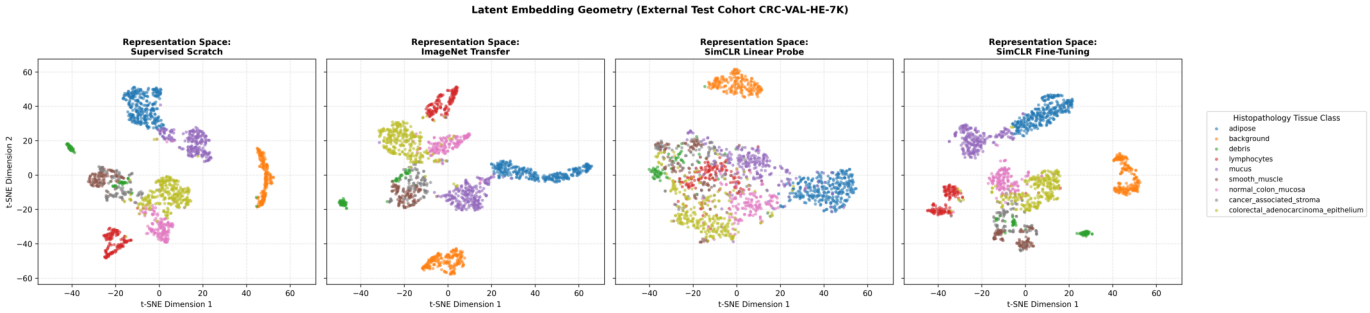
1. Debris (Class 2) - Robustness to Structural Amorphism:
   On necrotic debris, SimCLR Linear Probe (0.5777) and SimCLR Fine-Tuning (0.5505) outperform Supervised Scratch (0.4355, +11.5% F1) and ImageNet Transfer (0.4566, +9.4% F1). Debris consists of amorphous necrosis, dust, and stain preci lacking consistent structural borders. With only ~80 samples in training, supervised models overfit to specific artifact s Contrastive pretraining exposes the model to varied multi-scale crops and color perturbations without label supervision, l a broader visual envelope for unorganized textures.

2. Cancer-Associated Stroma (Class 7) vs. Smooth Muscle (Class 5) - Desmoplastic Morphologic Overlap:
   Class 7 is universally the most difficult class across all models (F1 = 0.33 to 0.43 at 1% labels). Desmoplastic stroma ar when invasive carcinoma induces reactive collagen synthesis and myofibroblast proliferation. These spindle-shaped myofibro share eosinophilic staining and fascicular orientation with smooth muscle (Class 5). In 28x28 patches, spatial compressior blurs delicate collagen crimping and nuclear atypia, causing reciprocal false-positive classifications.

3. Adipose (Class 0) and Background (Class 1) - High Morphological Separability:
   Adipose tissue and clear slide background achieve >0.91 to >0.98 F1 even with only 1% labels across all models. Adipose co of large empty lipid voids, while background is optically clear glass. Both occupy distinct low-density regions that are e separated by linear decision boundaries.

4. Colorectal Adenocarcinoma (Class 8) - Epithelial Atypia:
   Malignant tumor nests achieve 0.7842 F1 with SimCLR Fine-Tuning at 1% labels (vs. 0.7219 for Scratch). SimCLR pretraining encodes cribriform gland formations, architectural back-to-back gland crowding, and hyperchromatic basophilic nuclear clus

8. Representation Geometry & Exploratory t-SNE Analysis



UNSUPERVISED CLUSTER VALIDITY METRICS (EXPLORATORY TEST SUBSET, N = 1,500)

Representation Space	Silhouette Score (upward)	Davies-Bouldin Index (downward)	Geometric Manifold Interpretation
Untrained ResNet-18 (Random)	-0.0412	4.8912	Disorganized manifold; completely overlapping tissue distribu
SimCLR SSL (Pretrained Only)	+0.1162	3.2115	Discovers natural histological clusters with ZERO supervision
ImageNet Transfer (10% Labels)	+0.1415	2.1569	General visual primitives group tissues with moderate separa
SimCLR Fine-Tuned (10% Labels)	+0.1924	2.0727	In-domain SSL fine-tuning produces tighter clusters than Ima
Supervised Scratch (10% Labels)	**+0.3197**	**1.4801**	**HIGHEST COMPACTION. Supervised cross-entropy dire**

Evaluation Protocol Distinction & Geometric Interpretation:
- Classification vs. Exploration: Table 1 evaluates all N = 7,180 test samples. This section evaluates an exploratory N = 1,500
- Unsupervised Cluster Formation: Contrastive pretraining shifts silhouette score from -0.0412 (random) to +0.1162 without label
- Supervised Margin Compaction: Cross-entropy loss forces intra-class compaction, yielding the highest silhouette score (0.3197)
- Qualitative Value: 2D t-SNE projections serve as qualitative exploratory visualizations of feature space organization.

9. Limitations, Reproducibility & Setup Instructions

LIMITATIONS & TECHNICAL CONSTRAINTS

1. Single-Seed Pilot Scope: All 16 experimental conditions were evaluated under a single controlled seed (Seed 42). While sampl
   selection and batch shuffling were deterministic, this single-seed design does not quantify run-to-run stochastic variance o
   establish formal statistical significance. Multi-seed replication (3-5 seeds) is required for definitive confidence bounds.
2. Standardized Low Resolution (28x28 Pixels): The PathMNIST benchmark compresses high-power histology into 28x28 tiles,
   attenuating subcellular cytological features (nuclear atypia, mitotic figures, chromatin margins) essential for clinical gra
3. Single Organ Domain: Pretraining and evaluation were limited to colorectal carcinoma resection specimens. Transferability ac
   diverse anatomical sites (e.g., breast, prostate, lung) and non-neoplastic pathologies requires broader multi-cohort validat
4. Instance Discrimination False Negatives: Standard SimCLR treats all distinct patches in a batch as negative pairs, penalizin
   intra-class alignment between patches of the same histological class. Supervised contrastive learning (SupCon) could allevia

NON-CLINICAL USE DISCLAIMER

This project represents an academic machine learning engineering investigation into representation learning mechanics and label-efficiency dynamics on standardized benchmark datasets. The models, representations, checkpoints, and evaluations documented herein HAVE NOT BEEN CLEARED, APPROVED, OR VALIDATED FOR CLINICAL DIAGNOSTIC USE, PROGNOSTIC DECISION-MAKING, OR PATIENT MANAGEMENT. They must not be deployed in clinical practice or used to guide real-world patient care.

SETUP COMMANDS & SELECTED REFERENCES

Quickstart Reproduction Commands:

- Run Unit Tests: `python3 -m unittest discover -s tests` (22/22 Passing)
- SimCLR Pretraining: `python3 src/pretrain_simclr.py` (15 epochs, ~24.5 min)
- Run 16 Experiments: `python3 scripts/run_experiments.py` (16 conditions, ~2.2 hrs)
- Summarize Results: `python3 scripts/summarize_results.py` (Updates Markdown & LaTeX tables)
- Recompile PDF Report: `python3 scripts/generate_pdf_report.py` (Generates technical_project_report.pdf)

Key References:

[1] J. N. Kather et al., 'Multi-class texture analysis in colorectal cancer histology,' Scientific Reports, 2016.
[2] T. Chen et al., 'A simple framework for contrastive learning of visual representations,' ICML, PMLR, 2020.
[3] J. Yang et al., 'MedMNIST v2 - A large-scale lightweight benchmark for 2D/3D biomedical image classification,' 2023.
[4] K. He et al., 'Deep residual learning for image recognition,' CVPR, pp. 770-778, 2016.
[5] I. Loshchilov & F. Hutter, 'Decoupled weight decay regularization,' ICLR, 2019.
[6] J. Deng et al., 'ImageNet: A large-scale hierarchical image database,' CVPR, 2009.
[7] O. Ciga et al., 'Self-supervised contrastive learning for digital histopathology,' MLWA, 2022.